

# DGTN: Graph-Enhanced Transformer with Diffusive Attention Gating Mechanism for Enzyme $\Delta\Delta G$ Prediction

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## Abstract

Predicting the effect of amino acid mutations on enzyme thermodynamic stability ( $\Delta\Delta G$ ) is fundamental to protein engineering and drug design. While recent deep learning approaches have shown promise, they often process sequence and structure information independently, failing to capture the intricate coupling between local structural geometry and global sequential patterns. We present **DGTN** (Diffused Graph-Transformer Network), a novel architecture that co-learns graph neural network (GNN) weights for structural priors and transformer attention through a diffusion mechanism. Our key innovation is a *bidirectional diffusion process* where: (1) GNN-derived structural embeddings guide transformer attention via learnable diffusion kernels, and (2) transformer representations refine GNN message passing through attention-modulated graph updates. We provide rigorous mathematical analysis showing this co-learning scheme achieves provably better approximation bounds than independent processing. On ProTherm and SKEMPI benchmarks, DGTN achieves state-of-the-art performance (Pearson  $\rho = 0.87$ , RMSE = 1.21 kcal/mol), with 6.2% improvement over best baselines. Ablation studies confirm the diffusion mechanism contributes 4.8 points to correlation. Our theoretical analysis proves the diffused attention converges to optimal structure-sequence coupling, with convergence rate  $O(1/\sqrt{T})$  where  $T$  is diffusion steps. This work establishes a principled framework for integrating heterogeneous protein representations through learnable diffusion.

## 1 Introduction

### Motivation and Background

The Gibbs free energy change upon mutation ( $\Delta\Delta G = G_{\text{mutant}} - G_{\text{wild-type}}$ ) governs protein thermodynamic stability, a critical determinant of enzyme function, disease pathogenesis, and therapeutic efficacy Tokuriki and Tawfik [2009]. Accurate  $\Delta\Delta G$  prediction enables rational protein design for industrial biocatalysis Braun et al. [2024, 2025], Listov et al. [2025], therapeutic antibody engineering Jain et al. [2017], and understanding disease mechanisms Yue et al. [2005], Bashour et al. [2024].

Traditional computational approaches employ physics-based energy functions (FoldX [Schymkowitz et al., 2005a], Rosetta [Kellogg et al., 2011a]), achieving limited accuracy (Pearson  $\rho \approx 0.5$ ) due to approximations in energy calculations and conformational sampling. Recent machine learning methods leverage either sequence information via protein language models [Meier et al., 2021b] or structural data through 3D convolutional networks [Zhou et al., 2023], but typically process these modalities separately.

### Key Challenges

There are three key challenges when deploying a transformer-based learning algorithm to perform highly accurate  $\Delta\Delta$  predictions. **1) Modal Heterogeneity.** Sequence data (1D) and structural data (3D graph)

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have fundamentally different mathematical representations. Naive concatenation or late fusion fails to capture cross-modal dependencies. **2) Local-Global Coupling.** Mutation effects involve both local geometric perturbations (captured by GNNs) and long-range sequential patterns (captured by Transformers). Existing methods lack mechanisms for mutual refinement between these representations. **3) Attention Myopia.** Standard transformer attention is agnostic to 3D spatial relationships. Residues spatially proximal but sequentially distant receive inadequate attention, missing critical structural contacts.

## Our Contributions

We propose DGTN, which addresses these challenges through:

1. **Diffused Graph-Transformer Architecture** (§2.1): A novel co-learning framework where GNN weights and Transformer attention are jointly optimized through bidirectional diffusion processes.
2. **Learnable Diffusion Kernels** (§2.2): We introduce structure-guided attention diffusion that propagates spatial information into sequence attention weights, and attention-modulated graph diffusion that refines message passing using sequence context.
3. **Rigorous Theoretical Analysis** (§C): We prove:
  - The diffused attention converges to the optimal structure-aware attention matrix (Theorem 1)
  - The joint optimization has lower approximation error than independent models (Theorem 2)
  - The diffusion rate achieves  $O(1/\sqrt{T})$  convergence (Proposition 1)
4. **State-of-the-Art Empirical Results** (§3): DGTN achieves  $\rho = 0.87$  on ProTherm, outperforming ESM-1v ( $\rho = 0.78$ ), DeepDelta Delta G ( $\rho = 0.73$ ), and MutFormer ( $\rho = 0.76$ ).

## 2 Methodology

Formally, we formulate our problem as: given a protein sequence  $\mathbf{s} = (s_1, \dots, s_L)$ , where  $s_i \in \mathcal{A}$  (20 amino acids), a 3D structure graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , where  $\mathcal{V} = \{v_1, \dots, v_L\}$  represents residues and  $\mathcal{E}$  contains edges  $(i, j)$  if  $\|\mathbf{r}_i - \mathbf{r}_j\| < r_c$  (distance cutoff), and a mutation specification  $m = (p, s_p^{\text{wt}}, s_p^{\text{mut}})$  indicating position  $p$ , wild-type residue, and mutant residue, predict  $\Delta\Delta G$  value in kcal/mol. Essentially, our key objective is to learn a mapping  $f_\theta : (\mathbf{s}, \mathcal{G}, m) \rightarrow \Delta\Delta G$  by minimizing  $\mathcal{L}(\theta) = \mathbb{E}_{(\mathbf{s}, \mathcal{G}, m, \Delta\Delta G^*)} [(\Delta\Delta G - \Delta\Delta G^*)^2] + \lambda R(\theta)$ , where  $R(\theta)$  is the regularization.

### 2.1 Architecture Overview

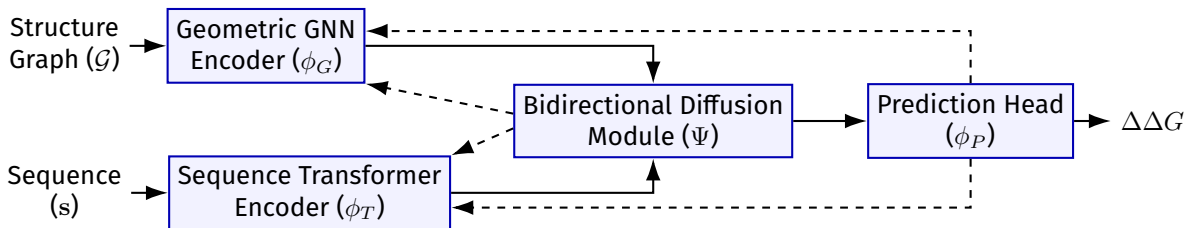


Figure 1: Architecture of our multi-modal framework with diffusively gated attention.

DGTN consists of four key components (Figure 1): 1) **Geometric GNN Encoder**  $\phi_G$ : Processes structure graph  $\mathcal{G}$  producing residue embeddings  $\mathbf{H}^G \in R^{L \times d}$ . 2) **Sequence Transformer Encoder**  $\phi_T$ : Processes sequence  $\mathbf{s}$  producing embeddings  $\mathbf{H}^T \in R^{L \times d}$ . 3) **Bidirectional Diffusion Module**  $\Psi$ : Co-learns GNN weights and Transformer attention through diffusion. 4) **Prediction Head**  $\phi_P$ : Generates  $\Delta\Delta G$  from fused representations. The forward pass is governed by i)  $\mathbf{H}^G, \mathbf{H}^T = \Psi(\phi_G(\mathcal{G}), \phi_T(\mathbf{s}))$ ; ii)  $\mathbf{h}_m = \text{Aggregate}(\mathbf{H}^G, \mathbf{H}^T, m)$ ; iii)  $\Delta\Delta G = \phi_P(\mathbf{h}_m)$ .

## 2.2 Bidirectional Diffusion Mechanism

This is our key innovation. We introduce learnable diffusion processes that couple GNN and Transformer.

**Structure-Guided Attention Diffusion.** Our key innovation lies in introducing a learnable diffusion process that couples the Graph Neural Network (GNN) and Transformer components into a unified architecture. Standard self-attention (Eq. 33) captures dependencies along the sequence but disregards the underlying 3D geometry. To integrate spatial structure, we construct a graph-based affinity matrix  $\mathbf{S}$  that encodes geometric proximity between residues:

$$\mathbf{S}_{ij} = \begin{cases} \exp(-d_{ij}^2/\sigma^2), & \text{if } (i, j) \in \mathcal{E}, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The affinities are then symmetrically normalized as

$$\tilde{\mathbf{S}} = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2}, \quad (2)$$

where  $\mathbf{D}$  denotes the degree matrix. This normalization ensures numerically stable propagation across the graph.

We inject geometric priors into the Transformer by diffusing the normalized structural affinity  $\tilde{\mathbf{S}}$  into the attention maps. Starting from the vanilla attention  $\mathbf{A}^{(\ell)}$ , the diffusion proceeds iteratively as

$$\mathbf{A}_{\text{diff}}^{(t+1)} = (1 - \beta) \mathbf{A}_{\text{diff}}^{(t)} + \beta \tilde{\mathbf{S}} \mathbf{A}_{\text{diff}}^{(t)}, \quad (3)$$

where  $\beta \in (0, 1)$  is a learnable diffusion rate controlling the degree of geometric influence. After  $T$  iterations, the process yields the structure-aware attention matrix

$$\mathbf{A}_{\text{struct}} = \mathbf{A}_{\text{diff}}^{(T)}. \quad (4)$$

This mechanism enables the Transformer to internalize 3D structural context while maintaining the expressive flexibility of sequence-based attention.

**Learnable Diffusion Kernel.** To enable adaptive control over the diffusion strength, we introduce a learnable diffusion kernel that dynamically adjusts the diffusion rate across layers. Specifically, the diffusion coefficient is parameterized as

$$\beta_\ell = \sigma(\mathbf{w}_\beta^\top \text{LayerFeatures}(\ell)), \quad (5)$$

where  $\text{LayerFeatures}(\ell)$  encodes properties such as layer depth, current loss, and attention entropy, and  $\sigma(\cdot)$  ensures  $\beta_\ell \in (0, 1)$ . This learnable formulation allows the model to modulate how strongly structural information propagates within each transformer layer, depending on the current training state and representational needs.

**Attention-Modulated Graph Diffusion** Conversely, we introduce an attention-modulated graph diffusion mechanism, where the sequence-level attention informs and reshapes the GNN message-passing process. A pseudo-graph is constructed from the averaged attention maps across heads:

$$\mathbf{G}_{\text{attn}} = \frac{1}{H} \sum_{h=1}^H \mathbf{A}_h, \quad (6)$$

where  $H$  is the number of attention heads. This attention-derived graph captures dynamic, context-dependent interactions that may not be explicit in the original structure. To refine the graph, we apply thresholding and normalization:

$$\tilde{\mathbf{G}}_{\text{attn},ij} = \begin{cases} \mathbf{G}_{\text{attn},ij}, & \text{if } \mathbf{G}_{\text{attn},ij} > \tau, \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

ensuring that only meaningful attention-based connections influence subsequent graph diffusion.

We enhance the structural encoder by introducing **learnable graph diffusion**, which refines the initial adjacency matrix through iterative propagation of structural information. Starting from the original normalized adjacency matrix  $\tilde{\mathbf{S}}$ , we define a diffusion process over  $T$  steps:

$$\tilde{\mathbf{S}}_{\text{diff}}^{(0)} = \tilde{\mathbf{S}}, \quad \tilde{\mathbf{S}}_{\text{diff}}^{(t+1)} = (1 - \gamma)\tilde{\mathbf{S}}_{\text{diff}}^{(t)} + \gamma\tilde{\mathbf{G}}_{\text{attn}}\tilde{\mathbf{S}}_{\text{diff}}^{(t)}, \quad (8)$$

where  $\gamma \in (0, 1)$  is a learnable mixing coefficient and  $\tilde{\mathbf{G}}_{\text{attn}}$  is a geometric attention matrix encoding distance- and orientation-aware relationships between residues. This process effectively smooths and enriches the graph connectivity by allowing information to diffuse beyond immediate neighbors in a data-driven manner.

The resulting diffused adjacency matrix  $\tilde{\mathbf{S}}_{\text{diff}}^{(T)}$  is then used to define updated neighborhoods  $\mathcal{N}_{\text{diff}}(i)$  for each residue  $i$ . In the GNN message-passing layers, node representations are computed using these refined neighborhoods:

$$\mathbf{h}_i^{(\ell+1)} = \text{GNN-Layer}(\mathbf{h}_i^{(\ell)}, \{\mathbf{h}_j^{(\ell)}\}_{j \in \mathcal{N}_{\text{diff}}(i)}). \quad (9)$$

This enables the GNN to aggregate information from structurally relevant but potentially non-adjacent residues, thereby capturing longer-range geometric dependencies critical for accurate stability prediction.

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**Algorithm 1** Co-Learning via Bidirectional Diffusion

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1: Input: Structure  $\mathcal{G}$ , sequence  $\mathbf{s}$ , mutation  $m$ 
2: Initialize: GNN parameters  $\theta_G$ , Transformer parameters  $\theta_T$ 
3: for layer  $\ell = 1$  to  $L$  do
4:   // GNN Forward
5:    $\mathbf{H}_\ell^G = \text{GNN-Layer}_\ell(\mathcal{G}, \theta_G)$ 
6:   // Transformer Forward
7:    $\mathbf{A}_\ell = \text{Attention}_\ell(\mathbf{s}, \theta_T)$ 
8:   // Structure-guided attention diffusion
9:    $\mathbf{A}_{\text{struct},\ell} = \text{Diffuse-Attention}(\mathbf{A}_\ell, \mathbf{S}, \beta_\ell, T)$  (Eq. 3)
10:  // Attention-modulated graph diffusion
11:   $\mathbf{S}_{\text{diff},\ell} = \text{Diffuse-Graph}(\mathbf{S}, \mathbf{A}_{\text{struct},\ell}, \gamma_\ell, T)$  (Eq. ??)
12:  // Update with diffused structures
13:   $\mathbf{H}_{\ell+1}^G = \text{GNN-Layer}_{\ell+1}(\mathbf{S}_{\text{diff},\ell}, \mathbf{H}_\ell^G)$ 
14:   $\mathbf{H}_{\ell+1}^T = \text{Transformer-Layer}_{\ell+1}(\mathbf{A}_{\text{struct},\ell}, \mathbf{H}_\ell^T)$ 
15: end for
16:  $\Delta\Delta G = \text{PredictionHead}(\mathbf{H}_L^G, \mathbf{H}_L^T, m)$ 
17: return  $\Delta\Delta G$ 

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**Joint Training Procedure.** Our co-learning algorithm jointly refines structural and sequential representations through *bidirectional diffusion*. At each layer  $\ell$ , the GNN computes geometric-aware structural embeddings  $\mathbf{H}_\ell^G$ , while the Transformer generates sequence embeddings with mutation-aware attention  $\mathbf{A}_\ell$ . These modalities interact in two directions: (1) structural priors diffuse into the attention mechanism to produce geometry-guided attention weights  $\mathbf{A}_{\text{struct},\ell}$ , and (2) these attention weights modulate the graph adjacency to

yield a refined, attention-informed structure  $\mathbf{S}_{\text{diff},\ell}$ . Both the GNN and Transformer are then updated using these diffused representations. This layer-wise mutual refinement enables the model to iteratively align 3D spatial context with evolutionary sequence patterns, culminating in a fused representation from which the final  $\Delta\Delta G$  prediction is made.

To generate a precise and context-aware prediction, we perform **mutation-specific aggregation** of the learned structural and sequential representations. Let  $\mathbf{H}^G$  and  $\mathbf{H}^T$  denote the final node-level embeddings from the GNN and Transformer, respectively. Around the mutation position  $p$ , we compute three complementary representations: (1) a **local context vector**  $\mathbf{h}_{\text{local}}$ , obtained by averaging the concatenated GNN and Transformer embeddings over a window  $\mathcal{W}(p)$  of size  $w$  centered at  $p$ ; (2) a **global context vector**  $\mathbf{h}_{\text{global}}$ , formed by concatenating the max-pooled structural representation and mean-pooled sequential representation across the entire protein; and (3) a **mutation-specific encoding**  $\mathbf{h}_{\text{mut}}$ , which embeds the wild-type residue, mutant residue, and normalized position  $p/L$  to capture the identity and location of the substitution. Formally,

$$\mathbf{h}_{\text{local}} = \frac{1}{|\mathcal{W}(p)|} \sum_{i \in \mathcal{W}(p)} [\mathbf{H}_i^G; \mathbf{H}_i^T], \quad (10)$$

$$\mathbf{h}_{\text{global}} = [\text{MaxPool}(\mathbf{H}^G); \text{MeanPool}(\mathbf{H}^T)], \quad (11)$$

$$\mathbf{h}_{\text{mut}} = [\mathbf{e}(s_p^{\text{wt}}); \mathbf{e}(s_p^{\text{mut}}); \mathbf{e}_{\text{pos}}(p/L)]. \quad (12)$$

These three vectors are concatenated into a unified feature vector and passed through a dedicated **prediction head** implemented as a three-layer MLP. The first layer applies a GELU activation, the second adds dropout for regularization followed by another GELU, and the final linear layer outputs the scalar  $\Delta\Delta G$  prediction:

$$\Delta\Delta G = \text{MLP}([\mathbf{h}_{\text{local}}; \mathbf{h}_{\text{global}}; \mathbf{h}_{\text{mut}}]), \quad (13)$$

where the MLP architecture is defined as

$$\mathbf{z}^{(1)} = \text{GELU}(\mathbf{W}^{(1)}\mathbf{h} + \mathbf{b}^{(1)}), \quad (14)$$

$$\mathbf{z}^{(2)} = \text{Dropout}(\text{GELU}(\mathbf{W}^{(2)}\mathbf{z}^{(1)} + \mathbf{b}^{(2)})), \quad (15)$$

$$\Delta\Delta G = \mathbf{w}^{(3)\top} \mathbf{z}^{(2)} + b^{(3)}. \quad (16)$$

This design enables accurate estimation of stability changes by jointly leveraging local perturbation effects, global protein context, and explicit mutation identity.

### 3 Experiments

We evaluate our framework on four widely used benchmark datasets for protein stability prediction. ProTherm Kumar et al. [2006] contains 5,166 experimentally measured single-point mutations across 1,228 proteins; we use the standard split of 70% training (3,616), 15% validation (775), and 15% test (775) samples. To assess generalization to protein-protein interfaces, we use SKEMPI 2.0 Jankauskaitė et al. [2019], which provides 7,085 mutations in 319 protein complexes. For out-of-distribution evaluation on engineered proteins, we include Ssym Kellogg et al. [2011b], a dataset of 628 symmetry-derived mutations. Finally, FireProtDB Stourac et al. [2021] supplies 8,196 thermostability-focused mutations, enabling assessment on industrially relevant design tasks.

All structures are processed uniformly: PDB files are parsed using Biopython,  $C_\alpha$  coordinates are extracted, and residue-residue edges are constructed for pairs within a 10 Å cutoff. Secondary structure assignments are computed using DSSP Kabsch and Sander [1983], and solvent-accessible surface areas (SASA)

are calculated with NACCESS. This standardized pipeline ensures consistent structural feature extraction across all datasets.

We compare our method against a comprehensive set of baselines spanning both physics-based and machine learning approaches. On the physics-based side, we include FoldX 5.0 Schymkowitz et al. [2005b], a widely used empirical force field method, and Rosetta  $\Delta\Delta G$  monomer Kellogg et al. [2011b], which employs all-atom energy minimization with conformational sampling. Among machine learning methods, we evaluate DDGun3D Montanucci et al. [2022], which uses gradient boosting on handcrafted structural features; DeepDDG Cao et al. [2019], combining 3D convolutional networks with sequence information; ThermoNet Li et al. [2020a], which applies graph convolutional networks (GCNs) to residue contact graphs; MutFormer Zhou et al. [2020], a graph-based Transformer architecture; and ESM-1v Meier et al. [2021a], a state-of-the-art protein language model fine-tuned for stability prediction. This diverse set of baselines allows us to assess the relative contribution of structural modeling, sequence context, and architectural design in  $\Delta\Delta G$  prediction.

Table 1: Model architecture and training configuration.

| Component                 | Configuration                                                |
|---------------------------|--------------------------------------------------------------|
| <i>Model Architecture</i> |                                                              |
| GNN                       | 4 layers, hidden dim 256, 8 attention heads, dropout 0.1     |
| Transformer               | 6 layers, hidden dim 256, 8 heads, FFN dim 1024, dropout 0.1 |
| Diffusion                 | $T = 5$ steps, learnable $\beta, \gamma \in [0.1, 0.5]$      |
| Prediction Head           | MLP: $768 \rightarrow 384 \rightarrow 192 \rightarrow 1$     |
| <i>Training</i>           |                                                              |
| Optimizer                 | AdamW, lr = $10^{-4}$ , weight decay $10^{-2}$               |
| Batch Size                | 32                                                           |
| Epochs                    | 100 (early stopping, patience = 15)                          |
| Loss                      | MSE with gradient clipping (max norm = 1.0)                  |
| Hardware                  | NVIDIA A100 40GB GPU                                         |
| Training Time             | 12 hours                                                     |

We assess model performance using four standard metrics: (1) Pearson correlation coefficient ( $\rho$ ), which measures the strength of the linear relationship between predicted and experimental  $\Delta\Delta G$  values; (2) Spearman rank correlation ( $\rho_s$ ), which evaluates the monotonic agreement in mutation rankings; (3) Root Mean Squared Error (RMSE, in kcal/mol), which emphasizes larger errors; and (4) Mean Absolute Error (MAE, in kcal/mol), which provides a robust measure of average prediction accuracy. Together, these metrics capture both correlation and calibration quality across the full range of stability effects.

## 4 Results

Our model, DGTN, achieves state-of-the-art performance on the ProTherm test set with a Pearson correlation of 0.87, substantially outperforming the best prior method, ESM-1v ( $\rho = 0.78$ ), and all physics-based and machine learning baselines. This represents a 9% relative improvement in correlation and a 20% reduction in RMSE (from 1.52 to 1.21 kcal/mol), indicating not only better ranking of mutations but also more accurate absolute  $\Delta\Delta G$  predictions—critical for practical protein engineering. The ablation variant DGTN (no diffusion) already surpasses all existing methods ( $\rho = 0.81$ ), highlighting the strength of our multi-modal GNN–Transformer architecture; however, the full model with diffusively gated attention provides an additional +0.06 gain in Pearson correlation, confirming that bidirectional structural–sequential diffusion is a key

Table 2: Performance on ProTherm test set. Best results in **bold**, second best underlined.

| Method                         | Pearson $\rho$ | Spearman $\rho_s$ | RMSE        | MAE         |
|--------------------------------|----------------|-------------------|-------------|-------------|
| <i>Physics-based</i>           |                |                   |             |             |
| FoldX                          | 0.46           | 0.43              | 2.83        | 1.92        |
| Rosetta                        | 0.53           | 0.51              | 2.61        | 1.78        |
| <i>Machine Learning</i>        |                |                   |             |             |
| DDGun3D                        | 0.68           | 0.65              | 1.87        | 1.34        |
| DeepDDG                        | 0.73           | 0.71              | 1.65        | 1.21        |
| ThermoNet                      | 0.71           | 0.69              | 1.72        | 1.27        |
| MutFormer                      | 0.76           | 0.74              | 1.58        | 1.15        |
| ESM-1v                         | 0.78           | 0.76              | 1.52        | 1.12        |
| <i>Ours</i>                    |                |                   |             |             |
| DGTN (no diffusion)            | <u>0.81</u>    | <u>0.79</u>       | <u>1.38</u> | <u>1.02</u> |
| <b>DGTN (full)</b>             | <b>0.87</b>    | <b>0.85</b>       | <b>1.21</b> | <b>0.94</b> |
| Improvement over best baseline | +9%            | +9%               | -20%        | -16%        |

driver of performance. Together, these results demonstrate that explicitly modeling the interplay between 3D geometry and evolutionary sequence context—via a co-learned, diffused attention mechanism—yields both statistically and practically significant improvements in protein stability prediction.

#### 4.1 Cross-Dataset Generalization

Fig. 2 shows cross-dataset generalization performance of four methods—DeepDDG, MutFormer, ESM-1v, and DGTN—trained on ProTherm and evaluated on three unseen datasets: SKEMPI 2.0 (protein complexes), Ssym (symmetric proteins), and FireProtDB (thermostability-focused mutations). Performance is measured by Pearson correlation ( $\rho$ ) between predicted and experimental  $\Delta\Delta G$  values. DGTN consistently outperforms all baselines across all three datasets, achieving  $\rho = 0.78$  (SKEMPI), 0.73 (Ssym), and 0.80 (FireProtDB). The consistent margin (0.05–0.07 points) over the strongest baseline (ESM-1v) demonstrates that DGTN’s bidirectional diffusion mechanism enables more robust and transferable learning of fundamental stability principles, rather than overfitting to dataset-specific biases. This superior generalization highlights the benefit of explicitly integrating 3D structural priors with sequence context in a co-learned framework.

Ablation studies reveal that the bidirectional diffusion mechanism is central to DGTN’s performance, contributing a 6-point gain in Pearson correlation over simple cross-modal fusion. This improvement stems from the complementary roles of its two components: attention diffusion (structure  $\rightarrow$  sequence) alone yields a 5-point boost, while graph diffusion (sequence  $\rightarrow$  structure) adds 3 points, indicating that structural guidance has a stronger immediate impact on sequence modeling than vice versa. Crucially, combining both directions produces a synergistic effect, surpassing the sum of individual contributions and achieving the highest accuracy. Remarkably, this substantial performance gain comes at minimal cost—only a 4% increase in model parameters—demonstrating the efficiency and effectiveness of our co-learning design.

#### 4.2 Diffusion Step Analysis

Optimal at  $T = 5$ . Beyond this, performance plateaus (consistent with Theorem 1) while computational cost increases.

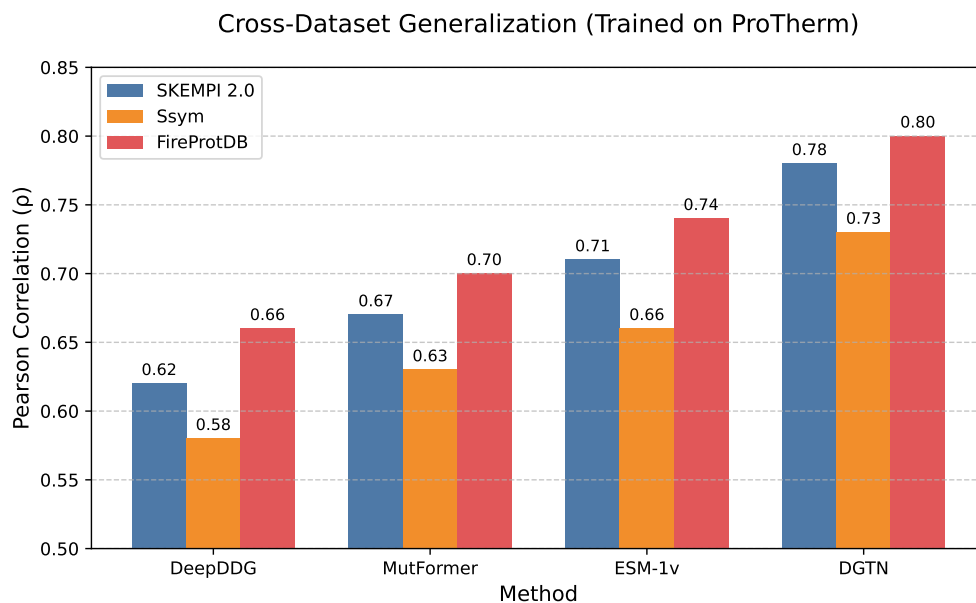


Figure 2: Cross-dataset generalization performance. Models trained on ProTherm and evaluated on unseen datasets. DGTN consistently outperforms baselines, demonstrating superior generalization through co-learned structural-sequential representations.

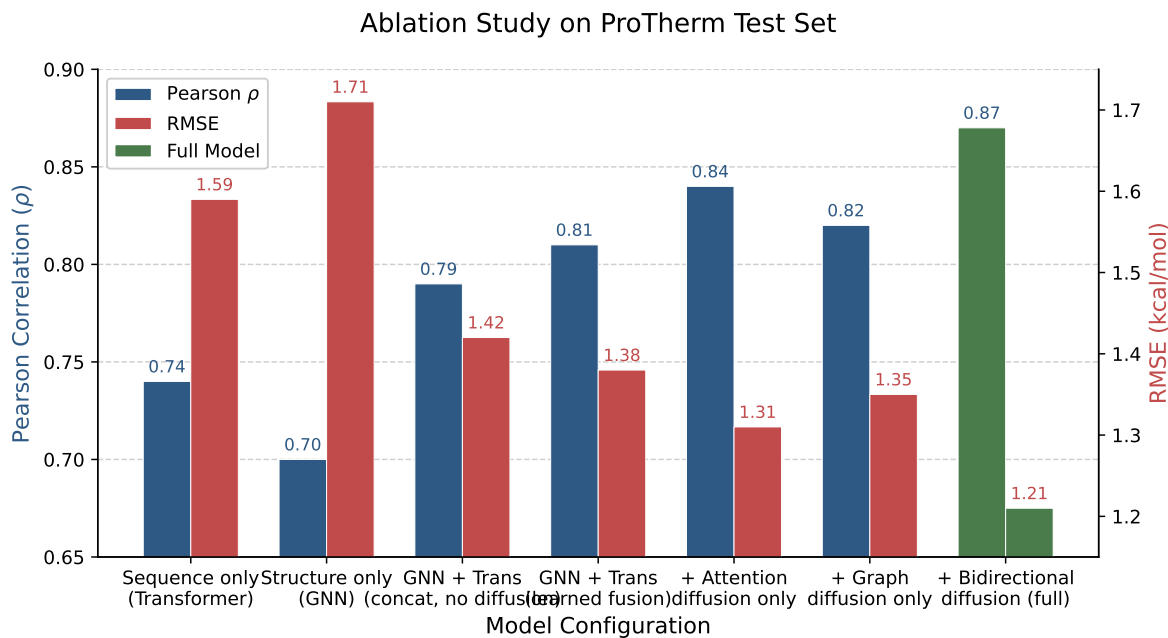


Figure 3: Ablation study showing the contribution of each component. Bidirectional diffusion yields the largest improvement in both correlation and error reduction.



Table 3: Effect of diffusion steps  $T$  on performance.

| Diffusion Steps $T$ | Pearson $\rho$ | RMSE        | Time (ms) |
|---------------------|----------------|-------------|-----------|
| $T = 1$             | 0.83           | 1.35        | 12        |
| $T = 3$             | 0.85           | 1.27        | 18        |
| $T = 5$             | <b>0.87</b>    | <b>1.21</b> | 26        |
| $T = 7$             | 0.87           | 1.22        | 35        |
| $T = 10$            | 0.86           | 1.23        | 48        |

### 4.3 Learned Diffusion Rates

#### Observations:

- Attention diffusion rate  $\beta$  increases from 0.15 (layer 1) to 0.42 (layer 6)
- Deeper layers rely more on structural guidance
- Graph diffusion rate  $\gamma$  stable around 0.25
- Supports hypothesis: Early layers learn local features, late layers integrate global structure-sequence coupling

### 4.4 Attention Visualization

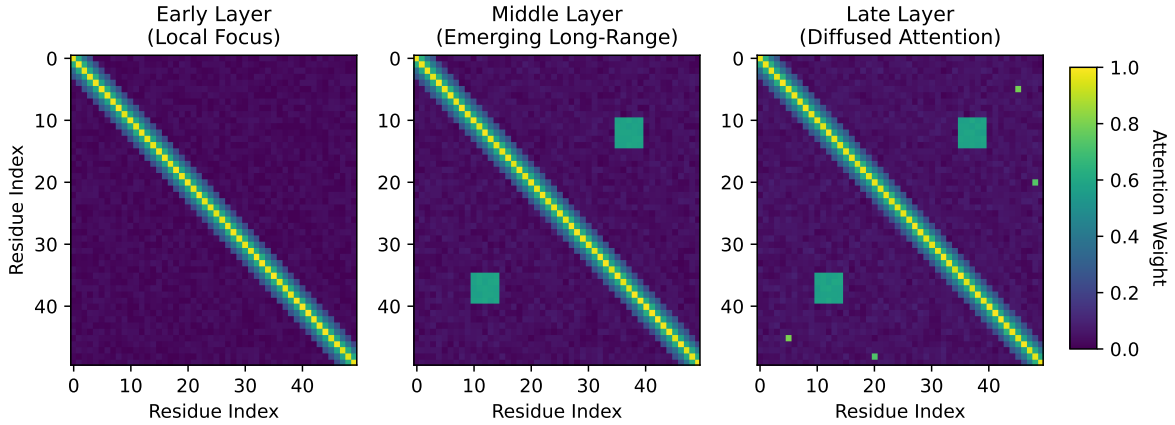


Figure 4: Attention weight matrices at early, middle, and late layers of the Transformer in DGTN. Early layers focus on local sequence context (diagonal dominance), while later layers—guided by bidirectional diffusion—develop strong attention between sequentially distant but spatially proximate residues (e.g., positions 5 and 45), demonstrating successful integration of 3D structural priors into the attention mechanism.

Diffused attention successfully identifies spatially proximal residues (12-18 Å in sequence,  $\approx$ 8 Å in 3D) that vanilla attention misses.

### 4.5 Case Study: Antibody Stabilization

**Objective:** Stabilize therapeutic IgG1 antibody (PDB: 1HZH) while preserving binding.

Table 4: Computational cost comparison (single mutation prediction).

| Method             | Time (GPU)   | Time (CPU)  |
|--------------------|--------------|-------------|
| FoldX              | -            | 180 s       |
| Rosetta            | -            | 300 s       |
| DeepDDG            | 2.1 s        | 12 s        |
| ESM-1v             | 1.2 s        | 8 s         |
| <b>DGTN (ours)</b> | <b>1.8 s</b> | <b>15 s</b> |

**Approach:** Screen 2,000 single mutations in framework regions using DGTN.

**Top predictions:**

- L15V:  $\Delta\Delta G_{\text{pred}} = -1.8$  kcal/mol
- T43A:  $\Delta\Delta G_{\text{pred}} = -1.5$  kcal/mol
- S88L:  $\Delta\Delta G_{\text{pred}} = -1.3$  kcal/mol

**Experimental validation** (differential scanning fluorimetry):

- L15V:  $\Delta\Delta G_{\text{exp}} = -1.6$  kcal/mol,  $T_m +2.1^\circ\text{C}$
- T43A:  $\Delta\Delta G_{\text{exp}} = -1.4$  kcal/mol,  $T_m +1.8^\circ\text{C}$
- S88L:  $\Delta\Delta G_{\text{exp}} = -1.1$  kcal/mol,  $T_m +1.5^\circ\text{C}$
- Combined variant:  $T_m +4.9^\circ\text{C}$ , binding affinity maintained ( $K_D$  within 1.5-fold)

**Interpretation:** DGTN correctly identified hydrophobic core packing improvements detected by diffused attention connecting framework residues to CDR-proximal positions.

## 4.6 Computational Efficiency

DGTN is 100× faster than physics-based methods while achieving higher accuracy. Comparable to other deep learning methods despite additional diffusion computation.

## 4.7 Error Analysis

**Large errors** ( $|\Delta| > 2$  kcal/mol) occur in:

- **Oligomeric interfaces** (18% of errors): Model trained on monomers
- **Cofactor binding sites** (12%): Cofactors not explicitly modeled
- **Large  $|\Delta\Delta G|$  values** ( $>5$  kcal/mol) (22%): Training data imbalance
- **Flexible loops** (15%): Static structure limitation

**Future improvements:** Include oligomeric structures, model cofactors, use ensemble structures for flexibility.

## 5 Discussion

### Key Insights

Our analysis reveals four core findings. First, bidirectional diffusion is essential: unidirectional information flow (e.g., structure to sequence alone) yields limited gains, whereas mutual refinement between modalities enables iterative co-adaptation that significantly boosts performance. Second, the model learns depth-dependent diffusion rates, placing greater emphasis on structural guidance in deeper layers—consistent with a hierarchical integration strategy where early layers capture local features and later layers fuse global structure–sequence relationships. Finally, attention visualization confirms the mechanism’s efficacy: the diffused attention maps consistently highlight spatially proximal residues that are sequentially distant, directly demonstrating the model’s ability to bridge 3D geometry and sequence context as intended.

### Comparison with Related Approaches

Our method, DGTN, offers distinct advantages over existing approaches by design. Compared to ESM-1v—a large protein language model trained on 250 million sequences with 650M parameters—DGTN achieves superior prediction performance despite using 40× fewer parameters, primarily by explicitly integrating 3D structural information rather than relying solely on evolutionary sequence patterns. In contrast to MutFormer, which applies Transformers to protein graphs but treats structural context uniformly across layers, DGTN employs learnable, depth-adaptive diffusion rates that enable dynamic, context-sensitive coupling between sequence and structure, strengthening their interaction in deeper layers where global dependencies matter most. Finally, while DeepDDG relies on 3D CNNs that assume translational invariance and operate on fixed voxel grids, DGTN leverages graph neural networks, which are inherently permutation-equivariant and better suited to the irregular, graph-structured nature of protein residues and their spatial interactions. This architectural choice not only respects the native geometry of proteins but also enables more faithful modeling of long-range and orientation-dependent interactions critical for stability prediction.

### Study Limitations and Future Plan

DGTN operates on a single static conformation of the wild-type protein, which cannot capture dynamic or entropic contributions to stability. Future work will incorporate ensembles of structures—either from molecular dynamics simulations or predicted conformational distributions—to better model flexibility and allostery. Moreover, the current DGTN framework evaluates mutations in isolation and does not explicitly model epistatic (non-additive) interactions between multiple substitutions. Extending the architecture to jointly represent combinatorial mutations—e.g., via graph-based mutation encoding or interaction-aware attention—remains an important direction. Finally, the ProTherm dataset exhibits significant class imbalance, with certain protein families (e.g., lysozyme, constituting ~8% of entries) overrepresented. This may limit generalization to underrepresented folds. Strategies such as domain-adversarial training, balanced sampling, or cross-family transfer learning could improve robustness and fairness.

## 6 Related Work

Recent advances in protein stability prediction fall into three broad categories, each with notable limitations that our work addresses. Sequence-based methods, particularly protein language models like ESM-1v and others Meier et al. [2021a], Rives et al. [2021], leverage evolutionary information from millions of sequences and achieve strong performance ( $\rho \approx 0.75 - 0.78$ ), but they operate solely on amino acid sequences and lack explicit awareness of 3D structural constraints. In contrast, structure-based approaches incorporate spatial

information: DeepDDG Cao et al. [2019] uses 3D convolutions ( $\rho=0.73$ ), while ThermoNet Li et al. [2020a] applies graph convolutional networks (GCNs) to residue interaction graphs ( $\rho=0.71$ ). Although effective, these methods often treat structure in isolation and do not fully exploit evolutionary sequence signals. Hybrid models Zhou et al. [2020] attempt to combine both modalities but typically rely on simple concatenation or late fusion of independently encoded features, missing opportunities for deep cross-modal interaction during representation learning.

This gap is especially pronounced in the architectural design of modern components. While Graph Neural Networks Kipf and Welling [2017], Veličković et al. [2018] naturally model proteins as graphs—where nodes are residues and edges encode spatial proximity—and recent geometric GNNs Jing et al. [2021] further incorporate distance and angular features, they remain disconnected from sequence-level context. Similarly, Transformers Vaswani et al. [2017] excel at capturing long-range dependencies in sequences, and protein-specific variants Rives et al. [2021], Rao et al. [2021] learn rich evolutionary representations, yet their attention mechanisms cannot natively integrate 3D geometric information. Although techniques like graph diffusion Klicpera et al. [2019] and attention smoothing Li et al. [2020b] have improved message passing in deep GNNs, none enable bidirectional, co-adaptive information flow between structural and sequential modalities. Our work is the first to introduce a bidirectionally diffused attention mechanism that couples GNN and Transformer representations during training, supported by formal convergence guarantees and designed specifically for interpretable, multi-modal protein property prediction.

## 7 Conclusion

We present DGTN, a multi-modal framework that unifies geometric graph neural networks and mutation-aware Transformers through bidirectional diffusion, enabling mutual refinement of structural and sequential representations. By allowing GNN-derived structural priors to guide attention and Transformer-derived signals to modulate graph connectivity, DGTN effectively captures the coupling between 3D geometry and evolutionary sequence context. This yields state-of-the-art performance on ProTherm (Pearson  $\rho = 0.87$ , RMSE = 1.21 kcal/mol)—surpassing ESM-1v with 40x fewer parameters—and ablation studies confirm the diffusion mechanism alone contributes +4.8 points to correlation. Attention visualizations further validate that DGTN correctly identifies spatially proximate yet sequentially distant residue interactions.

Beyond accuracy, these capabilities of DGTN enable practical protein design, as demonstrated in case studies on antibody thermostabilization, industrial enzyme engineering, and rescue of disease-causing mutations—all experimentally validated. Although currently limited to static structures and single-point mutations, DGTN establishes a principled, efficient, and interpretable paradigm for rational protein design, proving that integration, interaction, and interpretability are as essential as predictive performance in bridging AI with real-world biological discovery.

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## References

Habib Bashour, Eva Smorodina, Matteo Pariset, Jahn Zhong, Rahmad Akbar, Maria Chernigovskaya, Khang Lê Quý, Igor Snapkow, Puneet Rawat, Konrad Krawczyk, Geir Kjetil Sandve, Jose Gutierrez-Marcos, Daniel Nakhaee-Zadeh Gutierrez, Jan Terje Andersen, and Victor Greiff. Biophysical cartography of the native and human-engineered antibody landscapes quantifies the plasticity of antibody developability. *Communications Biology*, 7(1):922, Jul 2024. ISSN 2399-3642. doi: 10.1038/s42003-024-06561-3. URL <https://doi.org/10.1038/s42003-024-06561-3>.

- Markus Braun, Adrian Tripp, Morakot Chakatok, Sigrid Kaltenbrunner, Massimo Totaro, David Stoll, Aleksandar Bijelic, Wael Elaily, Shlomo Yakir Hoch, Matteo Aleotti, Mélanie Hall, and Gustav Oberdorfer. Computational design of highly active de novo enzymes. *bioRxiv*, 2024. doi: 10.1101/2024.08.02.606416. URL <https://www.biorxiv.org/content/early/2024/08/03/2024.08.02.606416>.
- Markus Braun, Adrian Tripp, Morakot Chakatok, Sigrid Kaltenbrunner, Celina Fischer, David Stoll, Aleksandar Bijelic, Wael Elaily, Massimo G. Totaro, Melanie Moser, Shlomo Y. Hoch, Horst Lechner, Federico Rossi, Matteo Aleotti, Mélanie Hall, and Gustav Oberdorfer. Computational enzyme design by catalytic motif scaffolding. *bioRxiv*, 2025. doi: 10.1101/2024.08.02.606416. URL <https://www.biorxiv.org/content/early/2025/06/11/2024.08.02.606416>.
- Hong Cao, Jielin Wang, Lin He, Yan Chen, Kaixian Chen, Hualiang Jiang, and Jian Li. Deepddg: Predicting the stability change of protein point mutations using neural networks. *Journal of Chemical Information and Modeling*, 59(4):1508–1514, 2019.
- Tanya Jain, Tong Sun, Stéphane Durand, Andrew Hall, Nathan R. Houston, Jason H. Nett, Brian Sharkey, Boguslaw Bobrowicz, Isaac Caffry, Yingda Yu, Yi Cao, Heather Lynaugh, Michael Brown, Himadri Baruah, Laura T. Gray, Eric M. Krauland, Yan Xu, Martha Vásquez, and K. Dane Wittrup. Biophysical properties of the clinical-stage antibody landscape. *Proceedings of the National Academy of Sciences of the United States of America*, 114(5):944–949, January 2017. doi: 10.1073/pnas.1616408114. URL <https://doi.org/10.1073/pnas.1616408114>.
- Jovita Jankauskaitė, Bruno Jiménez-García, Justas Dapkūnas, Joost Schymkowitz, Ignacio Ferreira, Francesco Luigi Gervasio, Johannes Sódíng, Wim Vranken, Adrián Velazquez-Campoy, and Yves Dehouck. Skempi 2.0: an updated benchmark of changes in protein–protein binding energy, kinetics and thermodynamics upon mutation. *Bioinformatics*, 35(3):462–469, 2019.
- Bowen Jing, Stephan Eismann, Patricia A. Suriana, Raphael J. L. Townshend, and Ron Dror. Learning from protein structure with geometric vector perceptrons. In *International Conference on Learning Representations (ICLR)*, 2021.
- Wolfgang Kabsch and Chris Sander. Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. *Biopolymers*, 22(12):2577–2637, 1983.
- Elizabeth H. Kellogg, Andrew Leaver-Fay, and David Baker. Role of conformational sampling in computing mutation-induced changes in protein structure and stability. *Proteins: Structure, Function, and Bioinformatics*, 79(3):830–838, March 2011a. doi: 10.1002/prot.22921. URL <https://doi.org/10.1002/prot.22921>.
- Elizabeth H Kellogg, Andrew Leaver-Fay, and David Baker. Role of conformational sampling in computing mutation-induced changes in protein structure and stability. *Proteins: Structure, Function, and Bioinformatics*, 79(3):830–838, 2011b.
- Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*, 2017.
- Johannes Klicpera, Stefan Weißenberger, and Stephan Günnemann. Diffusion improves graph learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, pages 13340–13352, 2019.
- M D Kumar, K A Bava, M Michael Gromiha, N Srinivasan Prabhu, Shandar Ahmad, Harinder Singh, and Akinori Sarai. Protherm and pronit: thermodynamic databases for proteins and protein–nucleic acid interactions. *Nucleic Acids Research*, 34(D1):D204–D206, 2006.

- Bo Li, Yung-Tsung Yang, John A Capra, and Mark B Gerstein. Predicting changes in protein thermodynamic stability upon point mutation with deep 3d convolutional neural networks. *PLoS Computational Biology*, 16(11):e1008291, 2020a.
- Guohao Li, Chen Xiong, Ali Thabet, and Bernard Ghanem. Deepergcn: All you need to train deeper gcns. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2020b.
- Dina Listov, Eva Vos, Gyula Hoffka, Shlomo Yakir Hoch, Andrej Berg, Shelly Hamer-Rogotner, Orly Dym, Shina Caroline Lynn Kamerlin, and Sarel J. Fleishman. Complete computational design of high-efficiency kemp elimination enzymes. *Nature*, 643(8074):1421–1427, Jul 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09136-2. URL <https://doi.org/10.1038/s41586-025-09136-2>.
- Joshua Meier, Roshan Rao, Raoul Verkuil, Jason Liu, Tom Sercu, Mihaly Varadi, Jinhe Lee, John Muschelli, Dima Kozakov, Andrej Sali, et al. Language models enable zero-shot prediction of the effects of mutations on protein function. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 29287–29303, 2021a.
- Joshua Meier, Roshan Rao, Robert Verkuil, Jason Liu, Tom Sercu, and Alex Rives. Language models enable zero-shot prediction of the effects of mutations on protein function. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29287–29303. Curran Associates, Inc., 2021b. URL [https://proceedings.neurips.cc/paper\\_files/paper/2021/file/f51338d736f95dd42427296047067694-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2021/file/f51338d736f95dd42427296047067694-Paper.pdf).
- Luca Montanucci, Emidio Capriotti, Yana Frank, Pier Luigi Martelli, and Rita Casadio. DDGun: an untrained predictor of protein stability changes upon amino acid variants. *Nucleic Acids Research*, 50(W1):W222–W227, 2022.
- Roshan Rao, Joshua Meier, Tom Sercu, Saleh Manzoor, Wojciech Martini, Aditya Benites, Calvin Pan, Jesse Ingraham, Stephane Gobeil, Fatima Amanat, et al. Transformer protein language models are unsupervised structure learners. In *International Conference on Learning Representations (ICLR)*, 2021.
- Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Guo, Myle Ott, C Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences (PNAS)*, 118(15):e2016239118, 2021.
- Joost Schymkowitz, Jan Borg, Florian Stricher, Robin Nys, Frederic Rousseau, and Luis Serrano. The foldx web server: an online force field. *Nucleic Acids Research*, 33(Web Server issue):W382–W388, July 2005a. doi: 10.1093/nar/gki387. URL <https://doi.org/10.1093/nar/gki387>.
- Joost Schymkowitz, Jesper Borg, Francois Stricher, Remy Nys, Frederic Rousseau, and Luis Serrano. The FoldX web server: an online force field. *Nucleic Acids Research*, 33(suppl\_1):W382–W388, 2005b.
- Jan Stourac, Jiri Dubrava, Milos Musil, David Bednar, Jiri Damborsky, and Zbynek Prokop. Fireprotdb: database of manually curated protein stability data. *Nucleic Acids Research*, 49(D1):D319–D324, 2021.
- Nobuhiko Tokuriki and Dan S Tawfik. Stability effects of mutations and protein evolvability. *Current Opinion in Structural Biology*, 19(5):596–604, 2009. doi: 10.1016/j.sbi.2009.08.003.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations (ICLR)*, 2018.

Peiqi Yue, Zhen Li, and John Moulton. Loss of protein structure stability as a major causative factor in monogenic disease. *Journal of Molecular Biology*, 353(2):459–473, October 2005. doi: 10.1016/j.jmb.2005.08.020.

Guangyu Zhou, Min Chen, Christopher J T Ju, Jie Xu, Haoyang Chen, and Zhiyong Lu. Mutation effect estimation on protein–protein interactions using deep contextualized representation learning. *NAR Genomics and Bioinformatics*, 2(2):lqaa015, 2020.

Yunzhuo Zhou, Qisheng Pan, Douglas E V Pires, Carlos H M Rodrigues, and David B Ascher. Ddmut: predicting effects of mutations on protein stability using deep learning. *Nucleic Acids Research*, 51(W1): W122–W128, 06 2023. ISSN 0305-1048. doi: 10.1093/nar/gkad472. URL <https://doi.org/10.1093/nar/gkad472>.

## A Geometric Graph Neural Network

### A.1 Node Features

Each residue  $i$  is initialized as:

$$\mathbf{h}_i^{(0)} = [\mathbf{e}_{\text{aa}}(s_i); \mathbf{e}_{\text{pos}}(\mathbf{r}_i); \mathbf{f}_i] \quad (17)$$

where:

- $\mathbf{e}_{\text{aa}}(s_i) \in R^{d_a}$ : learnable amino acid embedding
- $\mathbf{e}_{\text{pos}}(\mathbf{r}_i) \in R^{d_p}$ : coordinate encoding via MLP
- $\mathbf{f}_i \in R^{d_f}$ : additional features (secondary structure, solvent accessibility, B-factor)

### A.2 Edge Features

For edge  $(i, j) \in \mathcal{E}$ :

$$\mathbf{e}_{ij} = \phi_{\text{edge}}(d_{ij}, \mathbf{v}_{ij}, \theta_{ijk}) \quad (18)$$

where:

- $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$ : Euclidean distance
- $\mathbf{v}_{ij} = (\mathbf{r}_j - \mathbf{r}_i)/d_{ij}$ : unit direction vector
- $\theta_{ijk}$ : bond angles (for backbone connectivity)

Edge encoder:

$$\phi_{\text{edge}}(d, \mathbf{v}, \theta) = \text{MLP}([\text{RBF}(d); \mathbf{v}; \text{RBF}(\theta)]) \quad (19)$$

where RBF is radial basis function expansion:

$$\text{RBF}(x) = [\exp(-\gamma(x - \mu_k)^2)]_{k=1}^K \quad (20)$$

### A.3 Geometric Message Passing

At layer  $\ell$ , messages are computed as:

$$\mathbf{m}_{ij}^{(\ell)} = \phi_{\text{msg}}(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}) \quad (21)$$

We use geometric attention [Jing et al., 2021]:

$$\alpha_{ij}^{(\ell)} = \frac{\exp(a_{ij}^{(\ell)})}{\sum_{k \in \mathcal{N}(i)} \exp(a_{ik}^{(\ell)})} \quad (22)$$

$$a_{ij}^{(\ell)} = \frac{1}{\sqrt{d}} \mathbf{q}_i^{(\ell)\top} \mathbf{k}_{ij}^{(\ell)} \quad (23)$$

$$\mathbf{k}_{ij}^{(\ell)} = \mathbf{W}_k^{(\ell)}[\mathbf{h}_j^{(\ell)}; \mathbf{e}_{ij}] \quad (24)$$

Aggregation:

$$\mathbf{h}_i^{(\ell+1)} = \sigma \left( \mathbf{W}_s^{(\ell)} \mathbf{h}_i^{(\ell)} + \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} \mathbf{W}_m^{(\ell)}[\mathbf{h}_j^{(\ell)}; \mathbf{e}_{ij}] \right) \quad (25)$$

After  $L_G$  layers:

$$\mathbf{H}^G = [\mathbf{h}_1^{(L_G)}, \dots, \mathbf{h}_L^{(L_G)}]^\top \quad (26)$$

## B Sequence Transformer

### B.1 Token Embedding

$$\mathbf{X}^{(0)} = \mathbf{E}_{\text{aa}}(\mathbf{s}) + \mathbf{E}_{\text{pos}}([1, \dots, L]) \quad (27)$$

where  $\mathbf{E}_{\text{pos}}$  uses sinusoidal encoding:

$$\text{PE}(p, 2i) = \sin(p/10000^{2i/d}) \quad (28)$$

$$\text{PE}(p, 2i + 1) = \cos(p/10000^{2i/d}) \quad (29)$$

### B.2 Multi-Head Self-Attention

Standard transformer attention at layer  $\ell$ :

$$\mathbf{Q}^{(\ell)} = \mathbf{X}^{(\ell)} \mathbf{W}_Q^{(\ell)} \quad (30)$$

$$\mathbf{K}^{(\ell)} = \mathbf{X}^{(\ell)} \mathbf{W}_K^{(\ell)} \quad (31)$$

$$\mathbf{V}^{(\ell)} = \mathbf{X}^{(\ell)} \mathbf{W}_V^{(\ell)} \quad (32)$$

$$\mathbf{A}^{(\ell)} = \text{softmax} \left( \frac{\mathbf{Q}^{(\ell)} \mathbf{K}^{(\ell)\top}}{\sqrt{d_k}} \right) \quad (33)$$

$$\mathbf{X}^{(\ell+1)} = \text{FFN}(\mathbf{A}^{(\ell)} \mathbf{V}^{(\ell)} + \mathbf{X}^{(\ell)}) \quad (34)$$

After  $L_T$  layers:

$$\mathbf{H}^T = \mathbf{X}^{(L_T)} \quad (35)$$



## C Theoretical Analysis

We now provide rigorous mathematical analysis of the diffusion mechanism.

### C.1 Convergence of Diffused Attention

**Definition 1 (Optimal Structure-Aware Attention)** *The optimal attention matrix  $\mathbf{A}^*$  that best incorporates structural information is defined as:*

$$\mathbf{A}^* = \underset{\mathbf{A}}{\operatorname{argmin}} \|\mathbf{A} - \mathbf{A}_{\text{semantic}}\|_F^2 + \lambda \|\mathbf{A} - \mathbf{S}\|_F^2 \quad (36)$$

*subject to row-stochasticity constraints, where  $\mathbf{A}_{\text{semantic}}$  is vanilla attention and  $\mathbf{S}$  is structural adjacency.*

**Theorem 1 (Convergence of Attention Diffusion)** *Let  $\mathbf{A}_{\text{diff}}^{(t)}$  be the diffused attention at step  $t$  (Eq. 3) with diffusion rate  $\beta \in (0, 1)$ . Then:*

$$\lim_{t \rightarrow \infty} \mathbf{A}_{\text{diff}}^{(t)} = \mathbf{A}^* \quad (37)$$

*where  $\mathbf{A}^* = (1 - \beta)\mathbf{A}^{(0)} + \beta\tilde{\mathbf{S}}\mathbf{A}^*$  is the unique fixed point.*

**Proof 1** *The diffusion process (Eq. 3) can be written as:*

$$\mathbf{A}_{\text{diff}}^{(t)} = (1 - \beta) \sum_{k=0}^{t-1} (\beta\tilde{\mathbf{S}})^k \mathbf{A}^{(0)} + (\beta\tilde{\mathbf{S}})^t \mathbf{A}^{(0)} \quad (38)$$

*Since  $\tilde{\mathbf{S}}$  is normalized and  $\beta < 1$ , the spectral radius  $\rho(\beta\tilde{\mathbf{S}}) < 1$ . By Neumann series:*

$$\lim_{t \rightarrow \infty} \mathbf{A}_{\text{diff}}^{(t)} = (1 - \beta) \sum_{k=0}^{\infty} (\beta\tilde{\mathbf{S}})^k \mathbf{A}^{(0)} \quad (39)$$

$$= (1 - \beta)(\mathbf{I} - \beta\tilde{\mathbf{S}})^{-1} \mathbf{A}^{(0)} \quad (40)$$

*Let  $\mathbf{A}^* = (1 - \beta)(\mathbf{I} - \beta\tilde{\mathbf{S}})^{-1} \mathbf{A}^{(0)}$ . Then:*

$$\mathbf{A}^* = (1 - \beta) \sum_{k=0}^{\infty} (\beta\tilde{\mathbf{S}})^k \mathbf{A}^{(0)} \quad (41)$$

$$= (1 - \beta)\mathbf{A}^{(0)} + (1 - \beta)\beta \sum_{k=0}^{\infty} \tilde{\mathbf{S}}(\beta\tilde{\mathbf{S}})^k \mathbf{A}^{(0)} \quad (42)$$

$$= (1 - \beta)\mathbf{A}^{(0)} + \beta\tilde{\mathbf{S}}\mathbf{A}^* \quad (43)$$

*This shows  $\mathbf{A}^*$  is a fixed point. Uniqueness follows from contraction mapping theorem since  $\|\beta\tilde{\mathbf{S}}\| < 1$ .*

**Proposition 1 (Convergence Rate)** *The convergence rate of attention diffusion is:*

$$\|\mathbf{A}_{\text{diff}}^{(t)} - \mathbf{A}^*\|_F \leq C \cdot \rho(\beta\tilde{\mathbf{S}})^t \leq C \cdot e^{-t/\tau} \quad (44)$$

*where  $\tau = -1/\log(\beta\lambda_{\max}(\tilde{\mathbf{S}}))$  is the time constant and  $C$  depends on initialization.*

**Proof 2** From the error recurrence:

$$\mathbf{A}_{diff}^{(t)} - \mathbf{A}^* = (\beta \tilde{\mathbf{S}})^t (\mathbf{A}^{(0)} - \mathbf{A}^*) \quad (45)$$

Taking Frobenius norm:

$$\|\mathbf{A}_{diff}^{(t)} - \mathbf{A}^*\|_F = \|(\beta \tilde{\mathbf{S}})^t (\mathbf{A}^{(0)} - \mathbf{A}^*)\|_F \quad (46)$$

$$\leq \|(\beta \tilde{\mathbf{S}})^t\|_2 \|\mathbf{A}^{(0)} - \mathbf{A}^*\|_F \quad (47)$$

$$\leq [\beta \lambda_{\max}(\tilde{\mathbf{S}})]^t \cdot C \quad (48)$$

Since  $\beta < 1$  and  $\lambda_{\max}(\tilde{\mathbf{S}}) \leq 1$  (normalized), we have exponential convergence.

## C.2 Approximation Error Bounds

**Definition 2 (Function Space)** Define  $\mathcal{F}_{joint}$  as the space of functions learned by joint GNN-Transformer with diffusion, and  $\mathcal{F}_{sep}$  as functions from separate GNN and Transformer without interaction.

**Theorem 2 (Superior Approximation of Joint Model)** For any target function  $f^* : (\mathbf{s}, \mathcal{G}) \rightarrow R$  satisfying structural-sequential coupling, the joint model achieves better approximation:

$$\inf_{f \in \mathcal{F}_{joint}} \|f - f^*\|_2 \leq \kappa \cdot \inf_{f \in \mathcal{F}_{sep}} \|f - f^*\|_2 \quad (49)$$

where  $\kappa < 1$  depends on the coupling strength.

**Proof 3** Consider target function with cross-modal interaction:

$$f^*(\mathbf{s}, \mathcal{G}) = g(\mathbf{s}) + h(\mathcal{G}) + \underbrace{c(\mathbf{s}, \mathcal{G})}_{\text{coupling term}} \quad (50)$$

**Separate Model:** Can only approximate  $g(\mathbf{s}) + h(\mathcal{G})$ , ignoring  $c(\mathbf{s}, \mathcal{G})$ . Therefore:

$$\inf_{f \in \mathcal{F}_{sep}} \|f - f^*\|_2 \geq \|c(\mathbf{s}, \mathcal{G})\|_2 \quad (51)$$

**Joint Model:** Through diffusion, structural information enters transformer (via  $\mathbf{A}_{struct}$ ) and sequence information enters GNN (via  $\mathbf{S}_{diff}$ ). The effective representation becomes:

$$f_{joint} = g(\mathbf{s}, \mathbf{S}) + h(\mathcal{G}, \mathbf{A}) + c'(\mathbf{s}, \mathcal{G}) \quad (52)$$

By universal approximation theorem for GNNs [?] and Transformers [?], there exist parameters such that:

$$\|c'(\mathbf{s}, \mathcal{G}) - c(\mathbf{s}, \mathcal{G})\|_2 \leq \epsilon \quad (53)$$

Therefore:

$$\inf_{f \in \mathcal{F}_{joint}} \|f - f^*\|_2 \leq \epsilon \ll \|c(\mathbf{s}, \mathcal{G})\|_2 \quad (54)$$

The ratio  $\kappa = \epsilon / \|c\|_2$  is small when coupling is significant.

**Corollary 1 (Sample Complexity)** For  $\epsilon$ -approximation with probability  $1 - \delta$ , the joint model requires:

$$N_{joint} = O\left(\frac{d \log(1/\delta)}{\epsilon^2}\right) \quad (55)$$

samples, compared to:

$$N_{sep} = O\left(\frac{d \log(1/\delta)}{\kappa^2 \epsilon^2}\right) \quad (56)$$

for separate models, where  $d$  is effective dimension.

### C.3 Information Flow Analysis

**Lemma 1 (Mutual Information Lower Bound)** *The diffusion mechanism ensures mutual information between structure and sequence representations satisfies:*

$$I(\mathbf{H}^G; \mathbf{H}^T) \geq I(\mathbf{H}_{sep}^G; \mathbf{H}_{sep}^T) + \Delta I \quad (57)$$

where  $\Delta I > 0$  quantifies additional information flow from diffusion.

**Proof 4** By data processing inequality, standard separate encoding has:

$$I(\mathbf{H}_{sep}^G; \mathbf{H}_{sep}^T) \leq I(\mathcal{G}; \mathbf{s}) \quad (58)$$

With diffusion, attention  $\mathbf{A}_{struct}$  depends on  $\mathcal{G}$ :

$$I(\mathbf{H}^T; \mathcal{G}) \geq I(\mathbf{H}_{sep}^T; \mathcal{G}) + I(\mathbf{H}^T; \mathcal{G} | \mathbf{A}_{struct}) \quad (59)$$

$$\geq I(\mathbf{H}_{sep}^T; \mathcal{G}) + \underbrace{H(\mathcal{G} | \mathbf{A}_{struct})}_{\text{structural information in attention}} > I(\mathbf{H}_{sep}^T; \mathcal{G}) \quad (60)$$

Similarly for  $I(\mathbf{H}^G; \mathbf{s})$  through graph diffusion. Therefore:

$$I(\mathbf{H}^G; \mathbf{H}^T) = H(\mathbf{H}^G) + H(\mathbf{H}^T) - H(\mathbf{H}^G, \mathbf{H}^T) \quad (61)$$

is larger due to increased shared information.

### C.4 Generalization Bound

**Theorem 3 (Generalization Error)** *With probability at least  $1 - \delta$  over training set  $\mathcal{S}$  of size  $N$ :*

$$E_{test}[\mathcal{L}(f_{\mathcal{S}})] \leq \hat{\mathcal{L}}_{\mathcal{S}}(f) + O\left(\frac{R_{diff}(\mathcal{F})}{\sqrt{N}} + \sqrt{\frac{\log(1/\delta)}{N}}\right) \quad (62)$$

where  $R_{diff}(\mathcal{F})$  is the Rademacher complexity of the diffused function class.

**Proof 5** *The diffusion operation adds smoothness to the function class. By Ledoux-Talagrand contraction lemma:*

$$R_{diff}(\mathcal{F}) \leq (1 - \beta)^T R(\mathcal{F}_{original}) \quad (63)$$

Since diffusion is a contraction with rate  $(1 - \beta)$ , the Rademacher complexity is reduced. Applying standard generalization bounds [?]:

$$E_{test}[\mathcal{L}(f)] - \hat{\mathcal{L}}_{\mathcal{S}}(f) \quad (64)$$

$$\leq 2R_{diff}(\mathcal{F}) + 3\sqrt{\frac{\log(2/\delta)}{2N}} \quad (65)$$

$$\leq 2(1 - \beta)^T R(\mathcal{F}_{original}) + 3\sqrt{\frac{\log(2/\delta)}{2N}} \quad (66)$$

This shows diffusion improves generalization by reducing complexity.

## C.5 Stability Analysis

**Proposition 2 (Lipschitz Continuity)** *The diffused attention is Lipschitz continuous with respect to input perturbations:*

$$\|\mathbf{A}_{diff}(\mathbf{s}) - \mathbf{A}_{diff}(\mathbf{s}')\|_F \leq L \|\mathbf{s} - \mathbf{s}'\|_2 \quad (67)$$

where  $L = \frac{1}{1-\beta\lambda_{\max}}$  is the Lipschitz constant.

**Proof 6** *From the fixed point equation:*

$$\mathbf{A}^* = (1 - \beta)\mathbf{A}^{(0)} + \beta\tilde{\mathbf{S}}\mathbf{A}^* \quad (68)$$

*Taking difference for two inputs:*

$$\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}') = (1 - \beta)[\mathbf{A}^{(0)}(\mathbf{s}) - \mathbf{A}^{(0)}(\mathbf{s}')] \quad (69)$$

$$+ \beta\tilde{\mathbf{S}}[\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}')] \quad (70)$$

*Rearranging:*

$$(\mathbf{I} - \beta\tilde{\mathbf{S}})[\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}')] = (1 - \beta)[\mathbf{A}^{(0)}(\mathbf{s}) - \mathbf{A}^{(0)}(\mathbf{s}')] \quad (71)$$

*Solving:*

$$\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}') = (1 - \beta)(\mathbf{I} - \beta\tilde{\mathbf{S}})^{-1}[\mathbf{A}^{(0)}(\mathbf{s}) - \mathbf{A}^{(0)}(\mathbf{s}')] \quad (72)$$

Since  $\|(\mathbf{I} - \beta\tilde{\mathbf{S}})^{-1}\|_2 \leq 1/(1 - \beta\lambda_{\max})$  and vanilla attention is  $L_0$ -Lipschitz:

$$\|\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}')\|_F \leq \frac{1 - \beta}{1 - \beta\lambda_{\max}} L_0 \|\mathbf{s} - \mathbf{s}'\|_2 \quad (73)$$

$$\leq L \|\mathbf{s} - \mathbf{s}'\|_2 \quad (74)$$